

TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

**ENHANCING WI-FI SIGNAL COVERAGE USING
AN ADDITIONAL ROUTER**

BY

MA CHIT SU THWAY

VI IT - 1

GRADUATION THESIS

OCTOBER, 2025

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**A THESIS
SUBMITTED TO THE DEPARTMENT OF
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FOR THE DEGREE OF BACHELOR OF ENGINEERING
(INFORMATION TECHNOLOGY ENGINEERING)**

OCTOBER, 2025

THANLYIN

TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

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ABSTRACT

Wi-Fi connectivity has become an essential requirement for modern life, supporting activities such as web browsing, online communication, video streaming, and smart home device management. Despite its importance, many users encounter weak signals or dead zones where wireless coverage is insufficient. These issues are most commonly observed in larger homes, multi-story buildings, and structures with thick walls that obstruct signal transmission. Such limitations negatively impact the quality of network performance and reduce user efficiency. A significant component of this thesis involves the real-world implementation and empirical evaluation of a router-based Wi-Fi extension system. This thesis focuses on extending Wi-Fi coverage using an additional router as a cost-effective and practical approach. Unlike commercial range extenders, which may add unnecessary expenses, the use of a secondary router provides a flexible solution by repurposing existing or spare equipment. The research explores methods for deploying a secondary router to increase the range and stability of wireless signals within residential or building environments. The study investigates the common causes of weak Wi-Fi signals, including physical barriers, interference from other electronic devices, and poor router placement. Practical techniques for signal optimization, such as strategic positioning, configuration adjustments, and frequency band utilization, are also examined. The implementation of router-based Wi-Fi extension is tested in a real-world environment to evaluate improvements in coverage and stability. The overall aim of this thesis is to present an effective approach to mitigating Wi-Fi dead zones and enhancing network reliability. By applying the findings of this study, users can achieve improved connectivity in indoor environments without requiring high-cost infrastructure upgrades.

TABLE OF CONTENTS

		Page
	ACKNOWLEDGEMENT	i
	ABSTRACT	ii
	TABLE OF CONTENTS	iii
	LIST OF FIGURES	vi
	LIST OF TABLES	viii
CHAPTE R	TITLE	
1	INTRODUCTION	1
	1.1. Introduction to Enhancing Wi-Fi Signal Coverage Using Router	1
	1.2. Aim and Objectives of Thesis	2
	1.3. Scope of Thesis	2
	1.4. Outlines of Thesis	2
2	BACKGROUND THEORY	3
	2.1. Wireless Networks	3
	2.2. Wi-Fi Standards	4
	2.3. Wireless Signal Propagation	5
	2.4. Network Topologies for Coverage Extension	6
	2.4.1. Theoretical Calculation of Wi-Fi Coverage Range	6
	2.4.2. Theoretical Calculation of Wi-Fi Coverage Radius	7
	2.4.3. Theoretical Calculation of Wi-Fi Coverage Distance	7
	2.4.4. Calculation of Coverage	8
	2.5. IP Addressing Considerations	9
	2.5.1. Dynamic IP Address Assignment	9
	2.5.2. Advantages of Dynamic IP Addressing	9
	2.5.3. Static IP Address Assignment	10
	2.5.4. Advantages of Static IP Addressing	10
	2.5.5. Differences Between Dynamic and Static IP Assignment	11
	2.6. Modes of Operation in Router	12
	2.6.1. Access Point Mode	12
	2.6.2. Wireless Router Mode	12
	2.6.3. Range Extender mode or Repeater mode	13
	2.6.4. Bridge Mode	13

2.6.5. Client Mode	14
2.7. Signal Strength Basics	14
2.7.1. Signal Strength in Wireless Networks	14
2.7.2. RSSI in Wireless Networks	15
2.7.3. RSSI Versus dBm	15
2.7.4. Reading dBm	15
2.7.5. Ideal Signal Strength	16
2.8. Summary	17
3 DESIGN AND IMPLEMENTATION OF THE SYSTEM	18
3.1. Router Configuration	18
3.1.1. Overview of Router	18
3.1.2. Main Features	18
3.1.3. Front Panel Layout	18
3.1.4. Rear Panel	19
3.1.5. Connecting the Router	20
3.1.6. Login to Web-based Control Panel	21
3.1.7. Choosing Operation Mode	23
3.1.8. Network Configuration	23
3.1.9. Enabling Dual Band	24
3.1.10. Wireless 2.4GHz SSID Configuration	25
3.1.11. Wireless 2.4GHz Security Configuration	26
3.1.12. Wireless 5GHz SSID Configuration	27
3.1.13. Wireless 5GHz Security Configuration	28
3.1.14. DHCP Setting	29
3.1.15. DHCP Client List	30
3.1.16. System Reboot	30
3.1.17. Router Status	31
3.1.18. Connecting the Up-Link Cable to Router	32
3.1.19. Internet Reachability Testing	33
3.2 Implementing the Router to the Existing Infrastructure	34
3.2.1. Planning for Router Implementation	35
3.2.2. Network Cabling	35
3.2.3. Main Processes Involved in LAN Cabling	35
3.2.4. Materials Selection	37